ADS401 | GROUP J | PRESENTATION 3

LibAI Survey Statistical Results

Revised analysis of 94 eligible responses

Design	Eligible sample	Focus
Cross-sectional survey	n = 94	Use, experience and priorities

ADS401 Group J

8 November 2026

Revised to include task demand, LibAI evaluation, barriers, priorities and exploratory associations

Scope

This report analyses questionnaire data only. It does not report an AI comparison experiment because no experimental condition has been assigned or collected.

01 Executive summary and sample

The revised analysis covers the central questionnaire variables rather than only usage, satisfaction and one priority question. The sample remains a non-probability survey of XJTLU postgraduate students; results describe this analysed sample and do not estimate tool performance under controlled conditions.

Stage	Removed	Remaining
Raw respondent export	-	124
Did not consent (Q1)	6	118
Not an XJTLU postgraduate (Q2)	11	107
Logical conflicts after review	13	94

Final composition

- Current LibAI users: 20.
- Former LibAI users: 12.
- Never used LibAI: 62.
- Used some generative AI in the past three months: 90; never used: 4.

Corrected exclusion description

The six Q5/Q10 conflicts are more precisely described as respondents who reported never using generative AI in Q5 while reporting current LibAI use and LibAI tasks. Six further respondents selected both "no LibAI use in the past three months" and specific LibAI tasks in Q10. Two selected both "no facilitating factor" and a support option in Q24. One respondent met two rules, producing 13 unique exclusions.

Key findings

- Reading, summarizing and explaining papers (57/90) and finding or screening literature (56/90) are the two most common AI-assisted academic tasks.
- LibAI is perceived as moderately helpful for task completion, but user ratings are lower for source verification, retrieval coverage and workflow integration.
- Accuracy, database coverage and verifiable sources are the most common barriers; speed is a less common barrier.
- Verifiable sources (26/94) and accuracy (24/94) are the first two priorities, while speed is selected by only 6/94.
- The results support an experiment focused on answer quality, source verifiability and time to useful answer, rather than speed alone.

02 Research questions and analysis framework

Question	Purpose
RQ1	How do XJTLU postgraduate students use generative AI and LibAI for academic tasks?
RQ2	How do LibAI users evaluate task help, usability, source verification and workflow integration?
RQ3	Which barriers, concerns and improvement priorities do users and non-users report?
RQ4	Is overall satisfaction associated with reuse intention among previous LibAI users?

Analysis decisions

- Single-choice items are summarised as counts and percentages.
- Multiple-choice items are reported as the number selecting each option; percentages do not sum to 100%.
- Ordinal ratings are reported with mean, standard deviation, median, interquartile range and agreement categories.
- Q12, Q13 and Q27 treat 'not applicable' or 'unable to judge' as missing. They are not scored as zero.
- Q20 raw codes 1-11 are mapped to a 0-10 recommendation score.
- Q28 uses the full n = 94 as its denominator.
- Q6 uses n = 90 AI users; Q10 uses n = 20 current LibAI users; Q12-Q20 use n = 32 previous LibAI users; Q23-Q25 use n = 62 never-users.

Statistical tests

Analysis	n	Method	Role
Priority by LibAI use history	94	Chi-square; Cramer's V	Confirm difference between previous and never-users
Nine importance items	82 complete	Friedman test	Check whether the nine improvement items differ overall
Q12 task help vs Q13 conditions	32	Paired Wilcoxon	Exploratory comparison of two 1-5 composites
Q18 and Q19 association	32	Spearman; permutation / bootstrap	Exploratory only

Interpretation boundary

The survey cannot test whether one AI is more accurate, faster or more usable in an actual task. Those are perceptual and cross-sectional results. Any claim about performance requires the separate task-based experiment.

Software: Python, pandas, NumPy, SciPy and Matplotlib. Outputs were generated in Python, not SPSS.

03 Participation and AI use

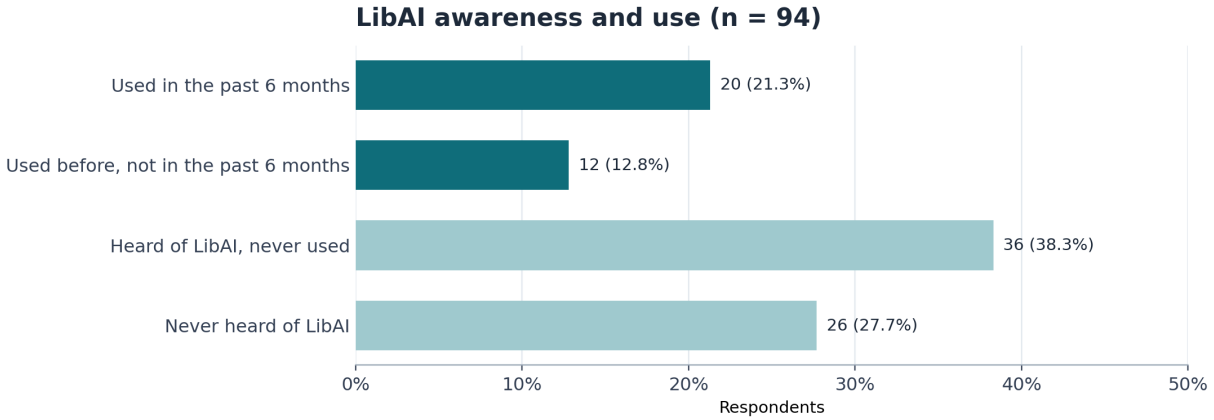


Figure 1. LibAI awareness and use. Q8, n = 94. Previous users combine current and former users.

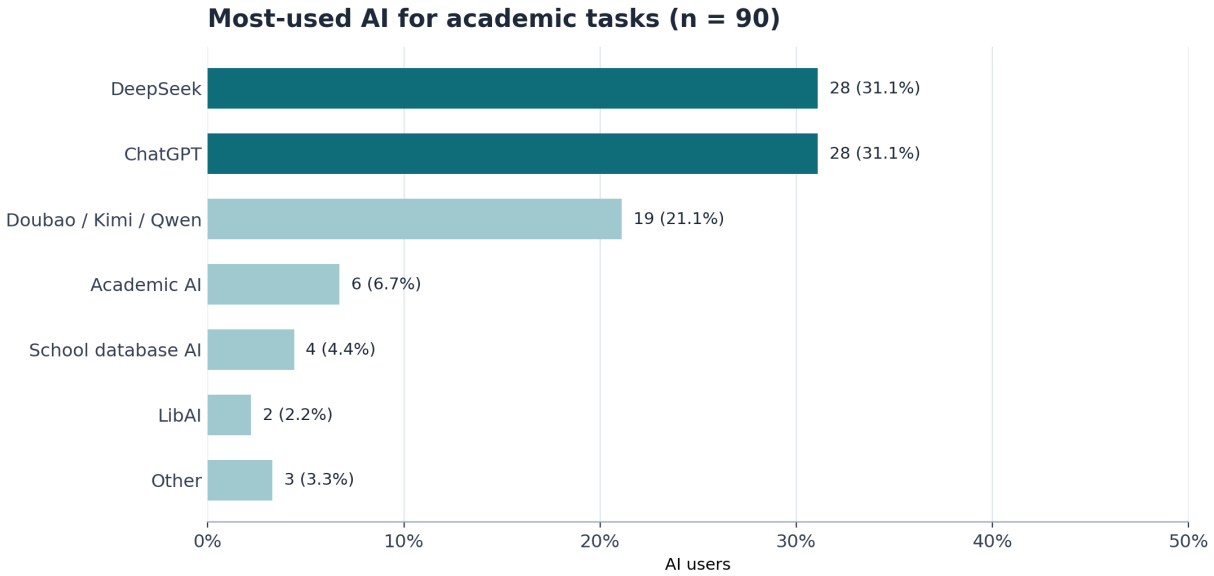


Figure 2. Most-used AI for academic tasks. Q7, n = 90 respondents who had used generative AI.

Implication

DeepSeek and ChatGPT each represent 28 of the 90 AI users. LibAI is selected as the most-used tool by only two respondents, so the experiment should include users and non-users rather than treating LibAI users as the default population.

Task-specific preference among previous LibAI users (n = 32)

Task	LibAI	External AI	Both	Not used
Clarify research questions / search terms	3 (9.4%)	6 (18.8%)	13 (40.6%)	10 (31.2%)
Find and screen literature	3 (9.4%)	7 (21.9%)	13 (40.6%)	9 (28.1%)
Read, summarize or explain papers	2 (6.2%)	8 (25.0%)	16 (50.0%)	6 (18.8%)
Compare papers / extract information	1 (3.1%)	7 (21.9%)	16 (50.0%)	8 (25.0%)
Obtain or format citations	2 (6.2%)	7 (21.9%)	15 (46.9%)	8 (25.0%)
Translate or read across languages	2 (6.2%)	7 (21.9%)	16 (50.0%)	7 (21.9%)

Q15, n = 32. 'Both' is the largest category for every task (40.6%-50.0%), while LibAI alone is selected by only 3.1%-9.4%. The two tools are generally treated as complementary rather than exclusive.

04 Task demand and LibAI use

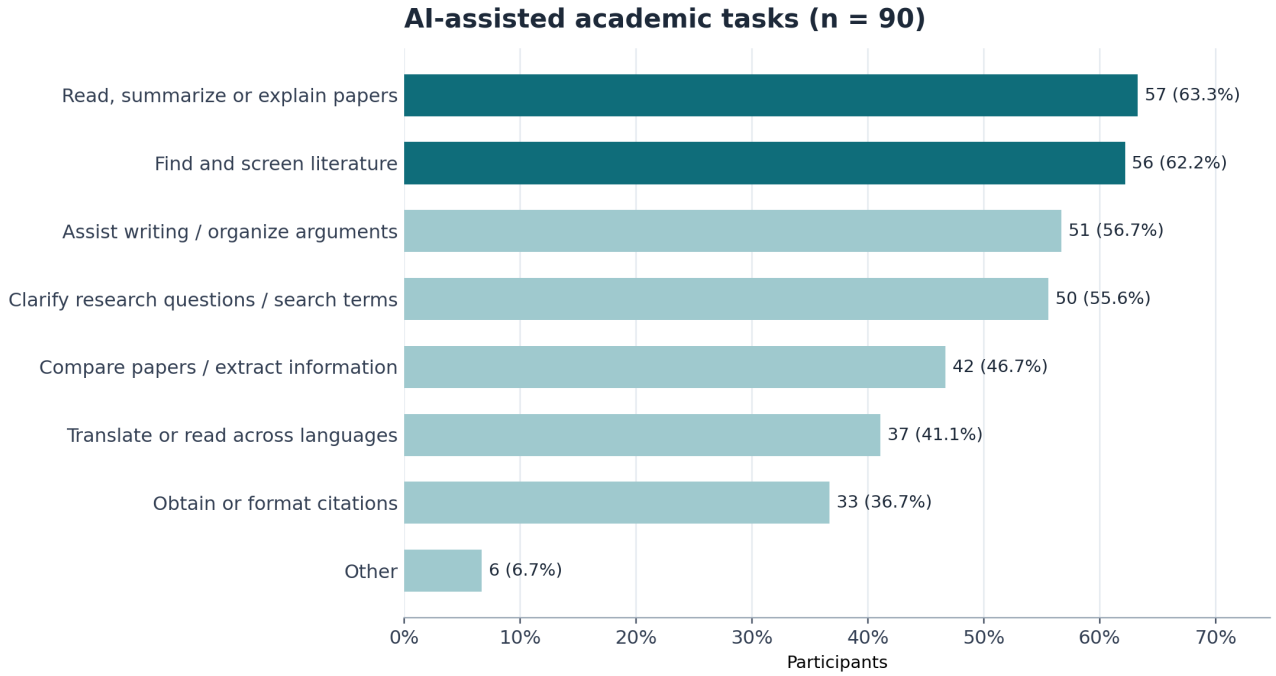


Figure 3. Academic tasks for which respondents used AI. Q6, n = 90 AI users; multiple selections are allowed.

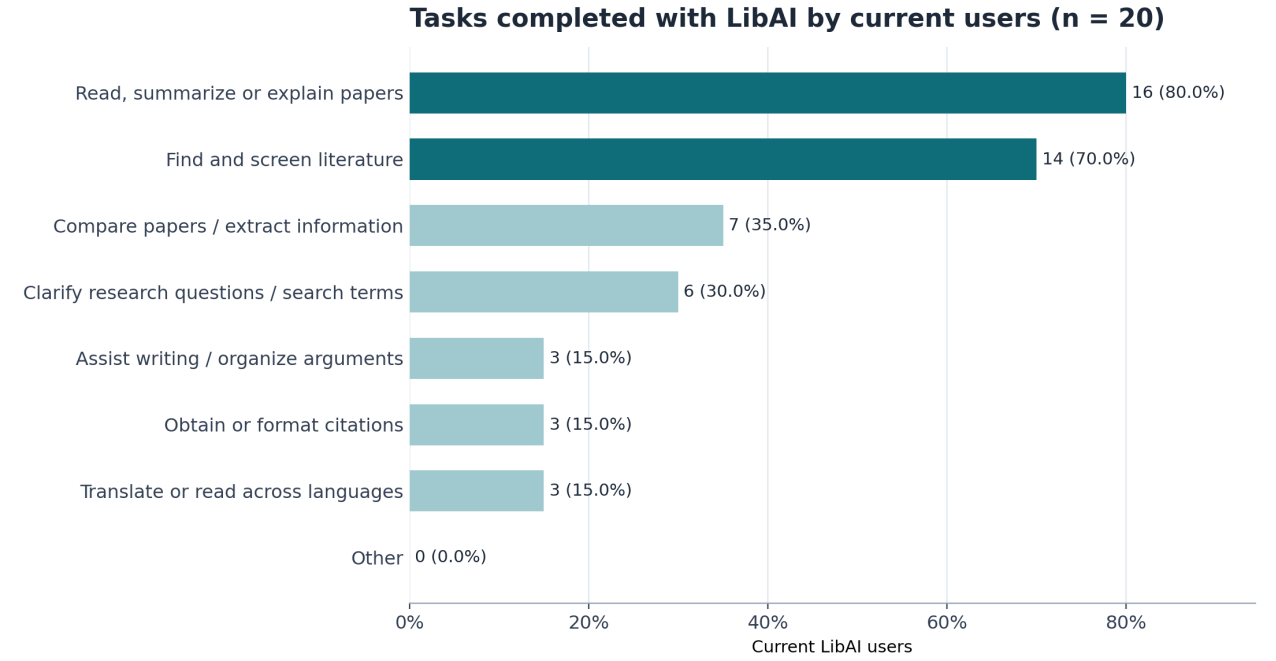


Figure 4. Tasks completed with LibAI among current users. Q10, n = 20; the 12 former users did not answer Q10 in this cleaned dataset.

Connection to experiment

The two most common task categories are reading/summarizing papers and finding/screening literature. These are therefore the appropriate fixed tasks for the subsequent experiment.

05 Perceived task help from LibAI

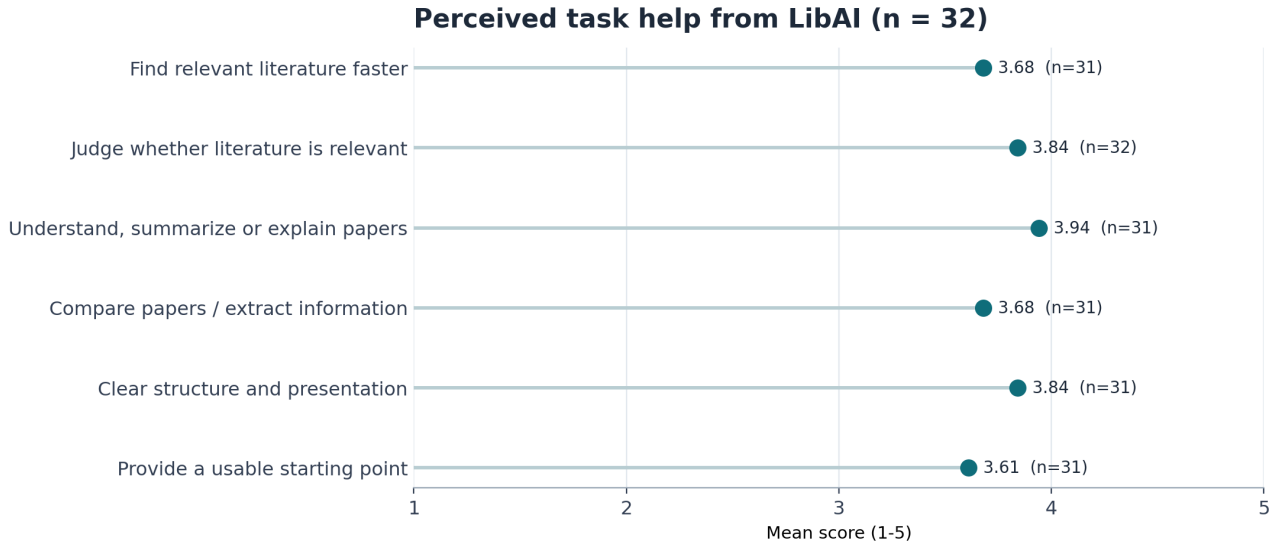


Figure 5. Perceived task help. Q12, previous LibAI users, n = 31-32 depending on applicability; 1 = strongly disagree, 5 = strongly agree. Cronbach's alpha = 0.90 for n = 31 complete responses.

Result	Value
Highest task-help mean	Understand/summarize papers: 3.94
Second-highest	Judge relevance and clear structure: 3.84 each
Lowest task-help mean	Usable starting point: 3.61
Agreement range	54.8% to 75.0% across the six items

Interpretation

LibAI is not evaluated as generally unhelpful. Users see moderate task value, especially for understanding papers and judging relevance. The weaker findings concern whether answers can be verified and incorporated into a dependable workflow.

06 Usability, trust and workflow conditions

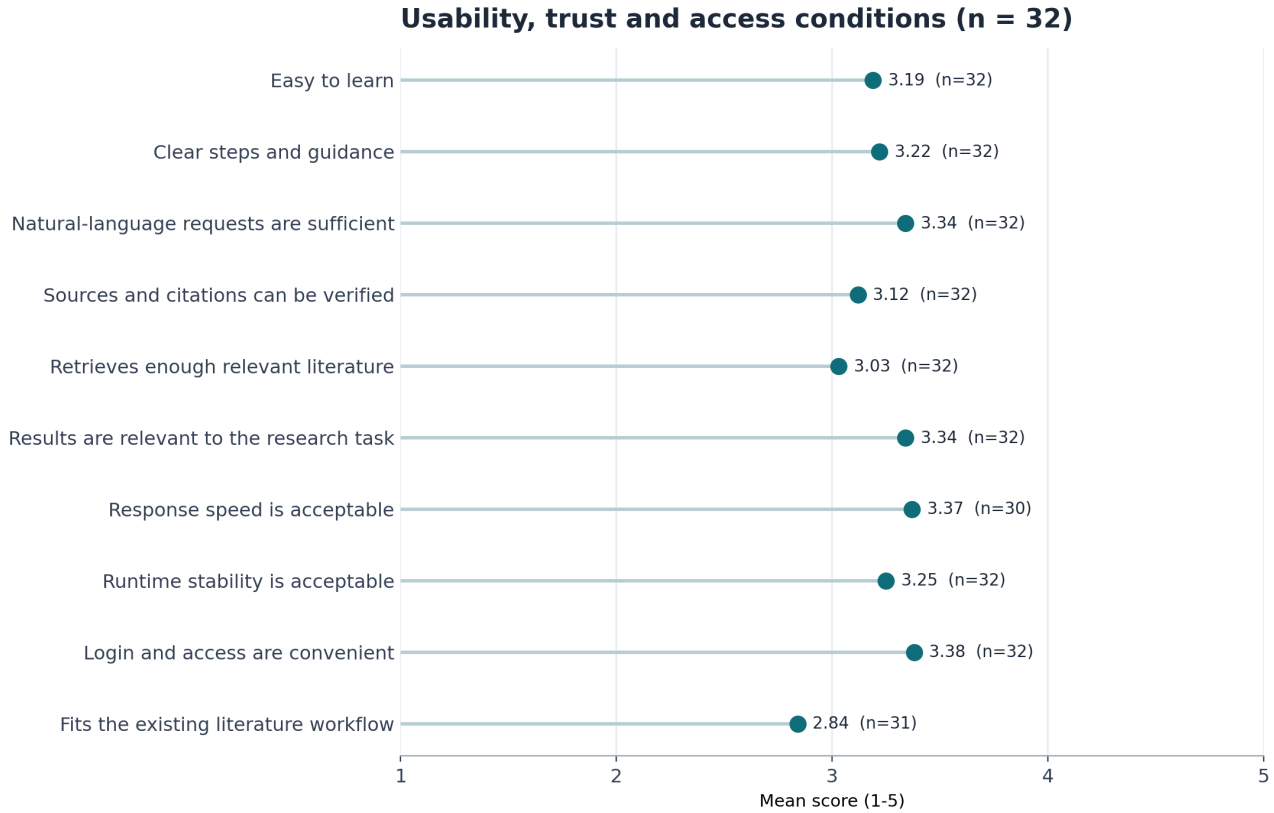


Figure 6. Usability, trust and access conditions. Q13, n = 30-32; 1 = strongly disagree, 5 = strongly agree. Items measuring these conditions have Cronbach's alpha = 0.90 among 30 complete responses.

Condition	Mean
Fits the existing literature workflow	2.84
Retrieves enough relevant literature	3.03
Sources and citations can be verified	3.12
Response speed is acceptable	3.37

Exploratory comparison

After taking within-person means, task help (3.72) is higher than the usability/trust composite (3.19), paired Wilcoxon p = 0.019, n = 32. This comparison is exploratory because the two blocks measure different constructs.

07 Concerns and stated barriers

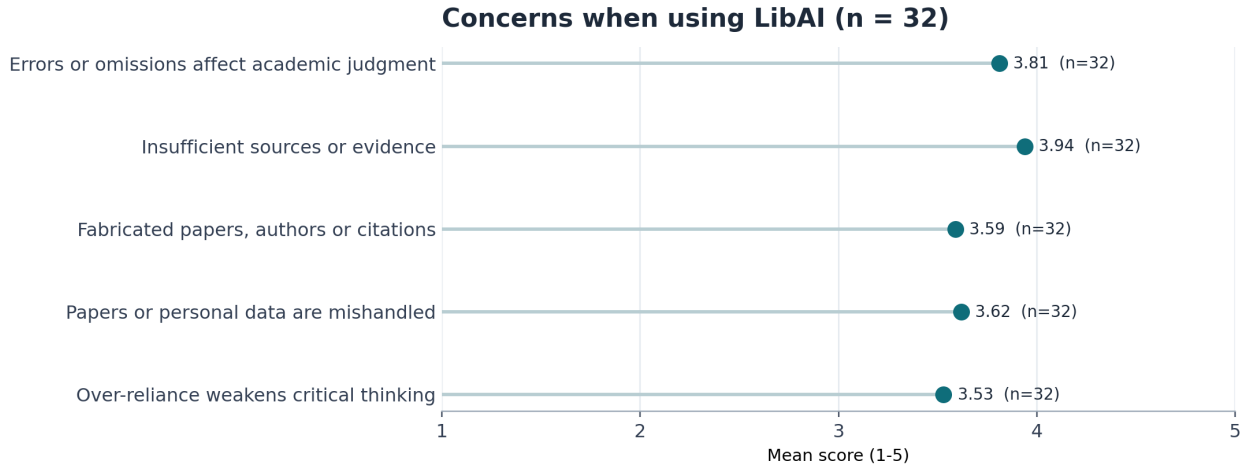


Figure 7. Concerns when using LibAI. Q14, n = 32; 1 = not at all concerned, 5 = very concerned. Cronbach's alpha = 0.91 for n = 32.

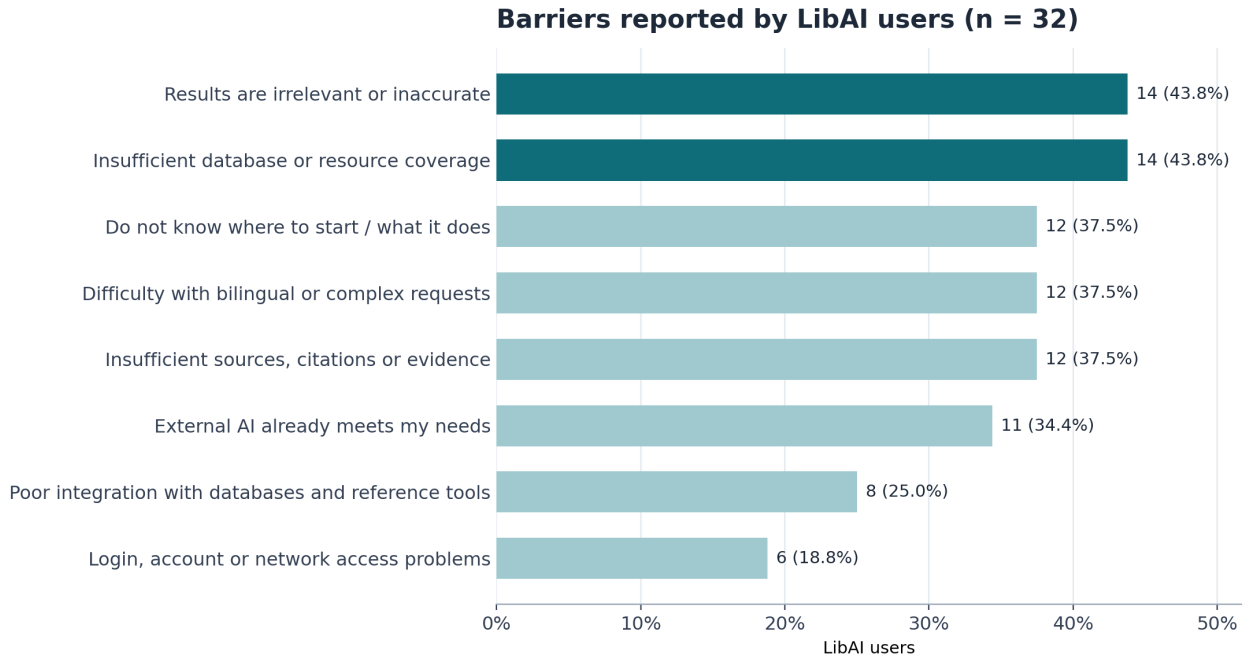


Figure 8. Barriers among previous LibAI users. Q16, n = 32; multiple selections are allowed. Percentages use 32 as the denominator.

Interpretation

Accuracy, database coverage and source/evidence quality are more frequently reported than speed or stability. The results support prioritising verified evidence and retrieval quality over latency alone.

08 Why non-users have not tried LibAI

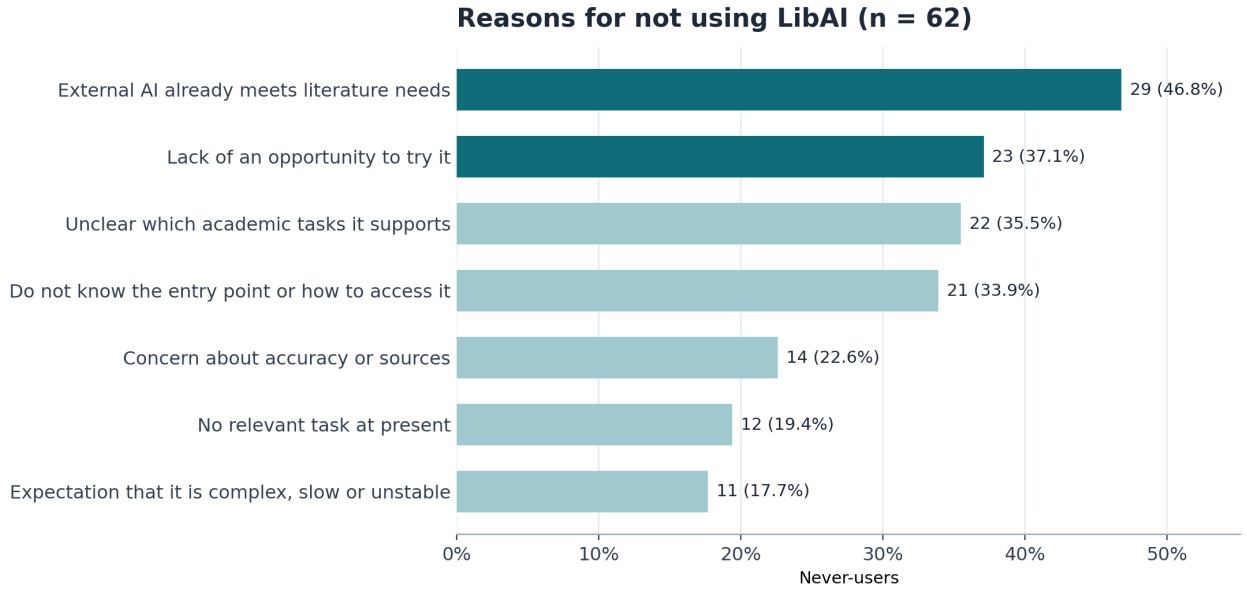


Figure 9. Reasons for not using LibAI. Q23, n = 62 never-users; multiple selections are allowed.

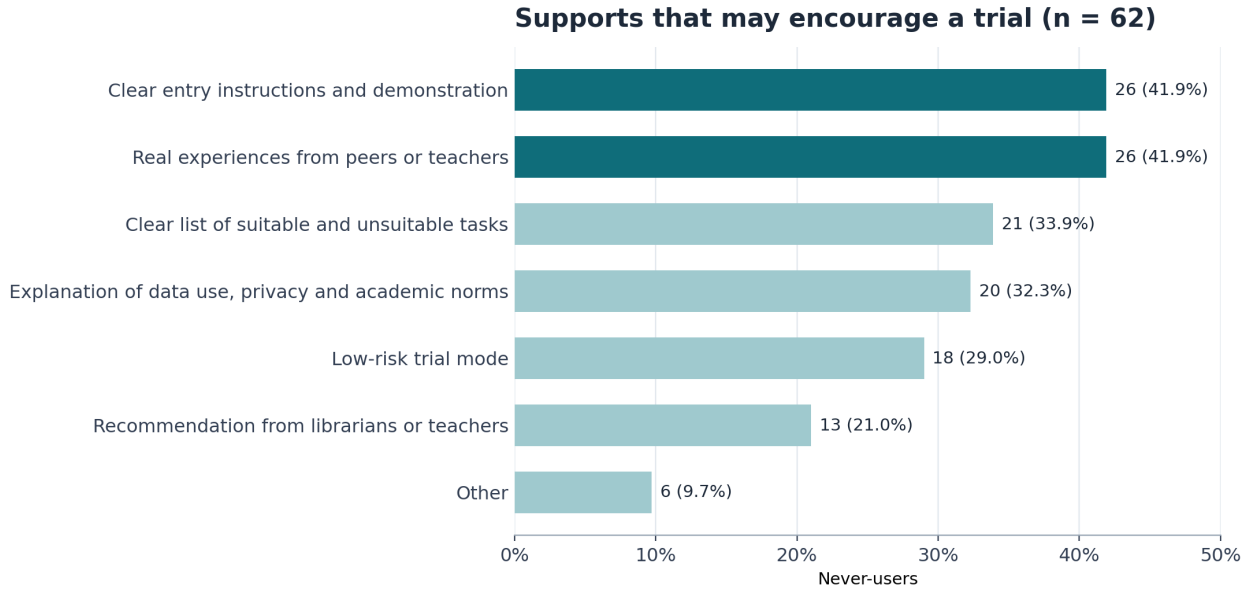


Figure 10. Supports that may encourage a trial. Q24, n = 62 never-users; multiple selections are allowed.

Key observation

Among non-users, 50.0% reported willingness to try LibAI if conditions improved (mean = 3.45/5, n = 62). The strongest access barriers are not primarily speed: they are task awareness, entry-point knowledge and the perception that existing AI already meets needs.

09 First priority and rated importance

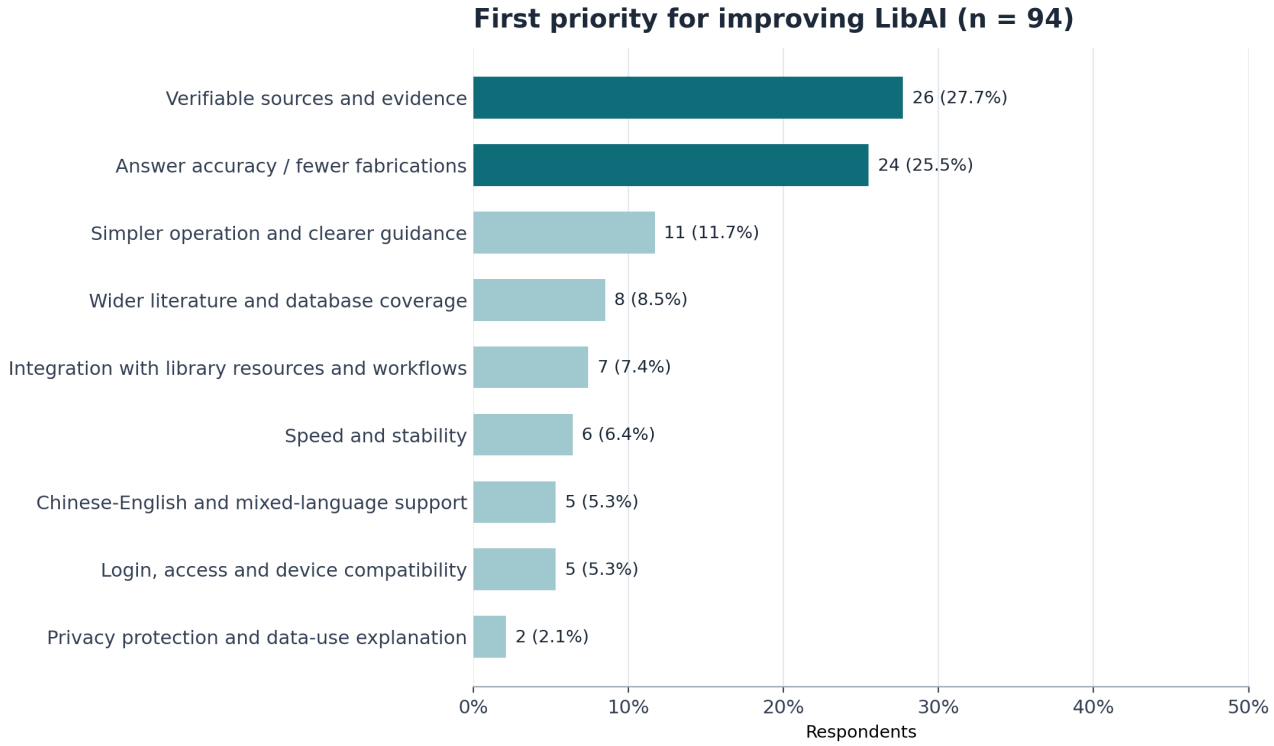


Figure 11. First improvement priority. Q28, n = 94, one option per respondent. Verifiable sources and accuracy account for 50/94 responses (53.2%).

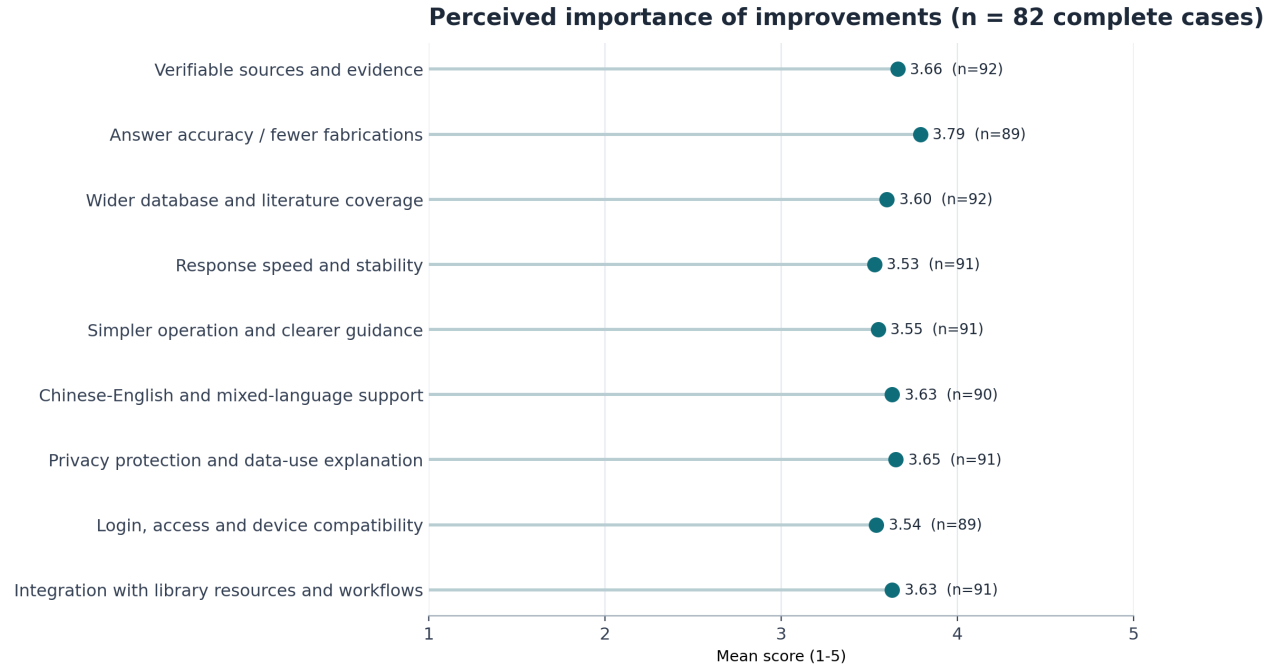


Figure 12. Mean importance of nine improvements. Q27, n = 89-92 depending on item; Friedman test across the complete cases, n = 82, chi-square(8) = 3.357, p = 0.910.

Why both questions are reported

All nine improvements received broadly high importance ratings. The absence of a significant overall ordering in Q27 makes the forced-choice Q28 useful: it shows which improvement respondents would address first.

10 User outcomes and exploratory association

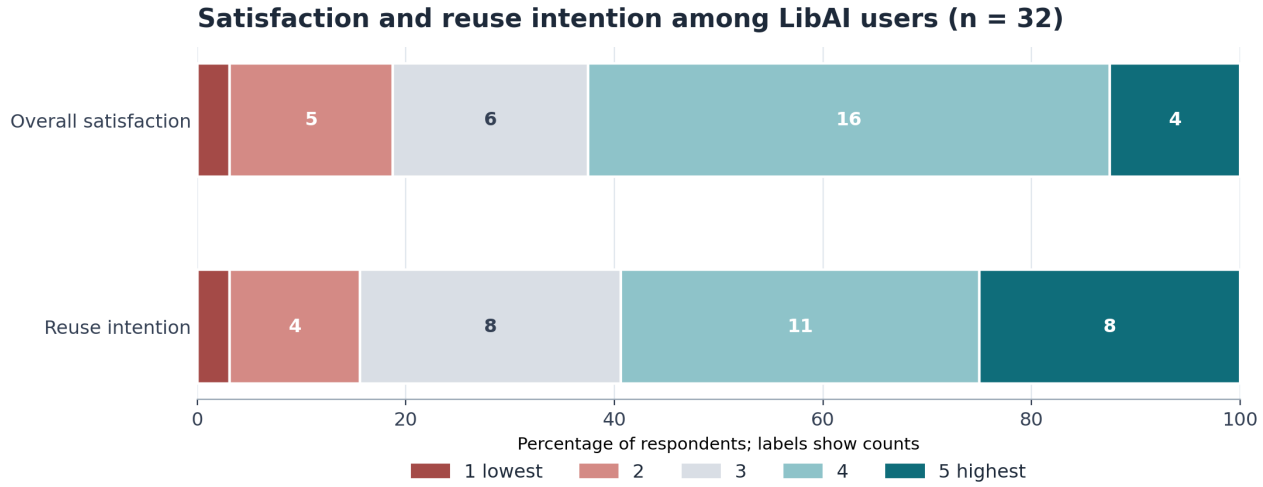


Figure 13. Satisfaction and reuse intention. Q18 and Q19, n = 32 previous LibAI users. Scores 4-5 represent 62.5% for satisfaction and 59.4% for reuse intention.

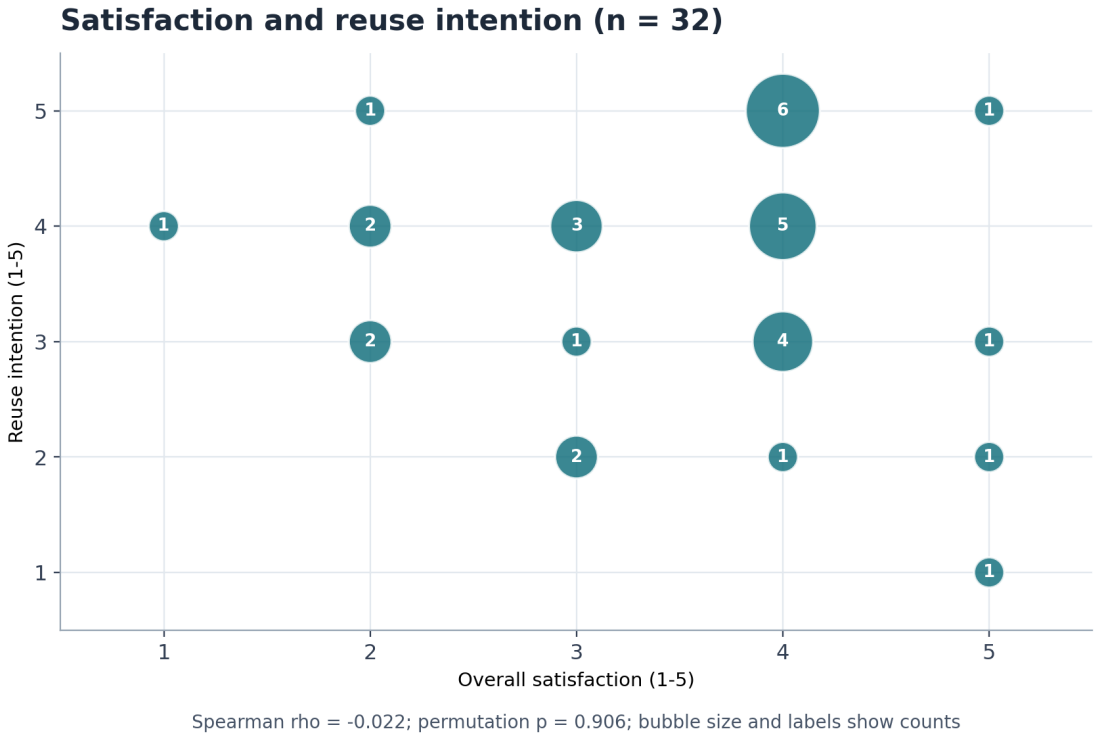


Figure 14. Response combinations for Q18 and Q19. The analysis is exploratory and does not establish causation.

Exploratory association

Spearman's rho = -0.022, permutation p = 0.906, 95% bootstrap interval [-0.40, 0.37]. There is insufficient evidence of a monotonic association in this small sample. The wide interval does not establish the absence of a relationship, and the near-zero estimate should not be used as a headline result.

Outcome summary

Outcome	Mean	Median	IQR
Overall satisfaction (Q18)	3.53	4.0	3.0-4.0
Reuse intention (Q19)	3.66	4.0	3.0-4.25
Recommendation, 0-10 (Q20)	4.72	5.0	NPS = -75.0

11 Discussion, implications and limitations

What the data show

- AI use is widespread, but LibAI is not the default tool: only 2 of 90 AI users name it as their most-used system.
- The dominant tasks are literature search and paper understanding, matching the two proposed experimental tasks.
- LibAI receives moderate task-help ratings but weaker ratings for source verification, sufficient retrieval and workflow integration.
- Users report concerns about source evidence and accuracy more often than about speed.
- Non-users are blocked more by awareness, entry points and the sufficiency of external AI than by speed alone.
- The first-priority results concentrate on verifiable sources and accuracy rather than latency.

Implications for the experiment

- Primary outcomes should include task success, answer quality, source verifiability and task completion accuracy.
- Time to useful answer should be measured as a secondary outcome, not the only convenience measure.
- LibAI should be compared with DeepSeek and ChatGPT on identical tasks with identical instructions.
- Users who have and have not used LibAI should both be recruited so familiarity does not determine the result.
- The interview should ask why a response is judged sufficient and how source checking changes trust.

Limitations

- The survey is cross-sectional and descriptive; it does not measure actual AI response quality or task completion time.
- The final 94 cases depend on logical-conflict exclusions decided after reviewing the data.
- The 32 previous LibAI users provide limited power for subgroup and correlation analyses.
- Ratings are self-reported and may reflect presentation, brand expectations and memory rather than independently measured performance.
- The survey does not include open-ended responses in this analysis dataset, so qualitative themes cannot be inferred from this file.

Main interpretation

The questionnaire is best used to identify the problem: perceived verification, accuracy, retrieval and workflow weaknesses. The task experiment is needed to test whether those perceived weaknesses appear in actual responses and how they relate to time to useful answer.

Evidence trail

Analysis data: 94-case workbook, variable dictionary and exclusion log. Question wording: LibAI participation questionnaire, Chinese revised version. The 13 exclusions are documented in the 94-case workbook; the report keeps the main analysis at n = 94 and does not apply the 107-case sensitivity analysis in this revision.

AI-use disclosure: OpenAI Codex assisted with data processing, statistical calculations, report drafting and formatting. ADS401 permits only non-substantive AI use; the group must confirm the permitted scope with the module leader and provide the required citation.